

Project Report: User Management REST API - Java Backend Developer

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Family:	Java Backend Developer
Level:	Beginner
Chapters:	6
Source:	CodeQuest project specification (official docs + industry patterns)

Summary

Spring Boot 3 CRUD API with JPA, validation, OpenAPI, and Docker Compose + PostgreSQL.

Chapters

Track: Project-Intro

Chapter 1: Project Overview [Beginner]

Chapter focus: Project Overview** *(Level: Beginner)

User Management REST API** is a beginner-level end-to-end project for Java Backend Developer. **Duration:** 2-3 weeks. **Spring Boot 3 CRUD API with JPA, validation, OpenAPI, and Docker Compose + PostgreSQL.** **Tools:** Java 21, Spring Boot 3, PostgreSQL, Docker.

Code Reference:

Objective: Design REST endpoints with proper HTTP status codes
Objective: Persist User entity with unique email
Objective: Write JUnit + MockMvc tests
Objective: Containerize with Flyway migrations

What it shows: Lists measurable outcomes before you start building.

Topic guide

What it actually is

A project overview defines scope, tools, and success criteria.

When to use it

At project kickoff.

Where to use it

Java Backend Developer portfolios and CodeQuest project submissions.

Real use example

A candidate lists this project on a Java Backend Developer resume with a demo link.

Track: Project-Phase

Chapter 2: Phase: Scaffold [Beginner]

Chapter focus: Phase: Scaffold** *(Level: Beginner)

Initialize Spring Boot with web, JPA, validation. Complete these tasks: Create User entity; Configure PostgreSQL. Deliverable for this phase: Health endpoint running.

Code Reference:

- [] Create User entity
- [] Configure PostgreSQL

What it shows: Checklist format helps track phase completion.

Topic guide

What it actually is

Phase Scaffold is one step in the full project lifecycle.

When to use it

During week 1 of the project timeline.

Where to use it

Java Backend Developer agile sprints and coursework milestones.

Real use example

Team demo shows the Health endpoint running to the teacher.

Chapter 3: Phase: REST Layer [Beginner]

Chapter focus: Phase: REST Layer** *(Level: Beginner)

Build controller, service, DTO layers. Complete these tasks: CRUD endpoints; Global exception handler. Deliverable for this phase: Postman collection all green.

Code Reference:

- [] CRUD endpoints
- [] Global exception handler

What it shows: Checklist format helps track phase completion.

Topic guide

What it actually is

Phase REST Layer is one step in the full project lifecycle.

When to use it

During week 2 of the project timeline.

Where to use it

Java Backend Developer agile sprints and coursework milestones.

Real use example

Team demo shows the Postman collection all green to the teacher.

Chapter 4: Phase: Testing [Beginner]**Chapter focus: Phase: Testing** *(Level: Beginner)**

Unit and integration tests. Complete these tasks: @WebMvcTest; @DataJpaTest. Deliverable for this phase: 80%+ coverage report.

Code Reference:

```
- [ ] @WebMvcTest
- [ ] @DataJpaTest
```

What it shows: Checklist format helps track phase completion.

Topic guide**What it actually is**

Phase Testing is one step in the full project lifecycle.

When to use it

During week 3 of the project timeline.

Where to use it

Java Backend Developer agile sprints and coursework milestones.

Real use example

Team demo shows the 80%+ coverage report to the teacher.

Chapter 5: Phase: Deploy [Beginner]**Chapter focus: Phase: Deploy** *(Level: Beginner)**

Docker Compose stack. Complete these tasks: Dockerfile; docker-compose.yml. Deliverable for this phase: README + demo video.

Code Reference:

```
- [ ] Dockerfile
- [ ] docker-compose.yml
```

What it shows: Checklist format helps track phase completion.

Topic guide

What it actually is

Phase Deploy is one step in the full project lifecycle.

When to use it

During week 4 of the project timeline.

Where to use it

Java Backend Developer agile sprints and coursework milestones.

Real use example

Team demo shows the README + demo video to the teacher.

Track: Project-Deliverables

Chapter 6: Final Deliverables and Review [Beginner]

Chapter focus: Final Deliverables and Review *(Level: Beginner)**

Submit all final deliverables: GitHub repo; OpenAPI spec; Test report; Demo video. Skills demonstrated: Spring Boot, JPA, REST, Docker. Prepare a 5-minute walkthrough video and README for CodeQuest teacher marking.

Code Reference:

```
Deliverable: GitHub repo
Deliverable: OpenAPI spec
Deliverable: Test report
Deliverable: Demo video
```

What it shows: Final checklist ensures nothing is missing at submission.

Topic guide**What it actually is**

Deliverable review confirms end-to-end project completion.

When to use it

Before deadline and teacher review.

Where to use it

CodeQuest project gallery and interviews.

Real use example

A student submits GitHub repo and passes the rubric.